

Kevin Xie

PHD CANDIDATE · PHYSICS · UNIVERSITY OF TORONTO

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Summary

PhD candidate in experimental quantum optics with 8+ years of experience spanning physics research, engineering, and industry R&D. Strong foundation in quantitative analysis, modeling, and development. Motivated by learning, adapting, and applying technical skills to new domains, including finance.

Experience

University of Toronto (Prof. Joseph Thywissen)

Toronto, Canada

PHD RESEARCHER, EXPERIMENTAL QUANTUM OPTICS

May 2020 – Present

- **Research focus:** precision measurements, high-dimensional data analysis, control of complex physical systems.
- Conducted experiments on ultracold quantum gases, including spectroscopy of strongly interacting fermions to study many-body phenomena.
- Applied model fitting, parameter estimation, and uncertainty analysis to extract physical parameters from noisy data.
- Deployed and maintained custom control, real-time monitoring, and automated data-acquisition systems (Python, C++).
- Led a research team of four, directing experimental strategy, coordinating measurement protocols, and managing data-analysis pipelines.
- Regularly communicated results to international collaborators; presented findings at conferences and peer-reviewed journals.

University of Waterloo (Prof. Jan Kycia)

Waterloo, Canada

UNDERGRADUATE RESEARCHER

Jan. – Sep. 2017, Sep. 2019 – Apr. 2020

- Engineered and analyzed thermal properties of components for a hybrid Helium-3 dilution refrigerator aimed at millikelvin-scale spin-qubit characterization.

Lumentum

Ottawa, Canada

OPTICAL ALGORITHMS DEVELOPER

Sep. – Dec. 2018, May – Aug. 2019

- Built a predictive failure-detection neural network model (>96% accuracy) to optimize calibration of LCoS components in optical networking devices, aimed at improving throughput and automation.
- Standardized test-station infrastructure across multiple facilities, eliminating bottlenecks and increasing throughput by up to 10%.
- Conducted comprehensive performance studies on next-generation bandwidth-enhancement algorithms for optical module R&D.
- Replaced outdated internal component data retrieval tool with modern Python-based solution adopted company-wide.

TRIUMF (T2K Canada)

Vancouver, Canada

UNDERGRADUATE RESEARCHER

Sep. 2017 – Apr. 2018

- Detected and corrected periodic temperature fluctuations that introduced systematic error in photomultiplier-tube characterization. Validated and reproduced by international collaborators (KEK).

Extracurricular Activities

Kaggle, Jane Street Group

Online

COMPETITION ENTRANT

2024

- Applied quantitative finance principles and machine learning algorithms (PyTorch, TensorFlow) to large-scale financial time-series data; developed feature engineering and model optimization with high-performance data tools (Polars).

University of Toronto Graduate Students' Union

Toronto, Canada

MEMBER OF BOARD OF DIRECTORS AND ELECTIONS & REFERENDA COMMITTEE

Oct. 2023 – Apr. 2024

- Organized and supported policy development and governance initiatives affecting 20,000+ graduate students, contributing to improvements in academic, financial, and community resources.

The 27th International Conference on Atomic Physics (ICAP 2022)

Toronto, Canada

VOLUNTEER

July 2022

- Assisted in organizing and running an international conference with 500+ participants, providing on-site logistics, A/V support, and management of virtual conference components.

Skills

Programming	Python (NumPy, SciPy, Pandas), MATLAB, Mathematica, C++, Linux, Git, LaTeX
Data	Uncertainty Analysis, Optimization, Signal Processing, Machine Learning (PyTorch, TensorFlow)
Languages	English (native), Cantonese (beginner/intermediate)
Interests	Music (Jazz), Boxing, Geopolitics, History

Education

University of Toronto

PHD PHYSICS

Toronto, Canada

Sep. 2020 - Present

University of Waterloo

BSC PHYSICS, CO-OP

Waterloo, Canada

Sep 2015 - Apr 2020

Publications

* indicates shared first authorship.

PUBLISHED

- C. J. Dale*, **K. G. S. Xie***, K. Pond Grehan, S. Zhang, J. Maki, and J. H. Thywissen. Emergent s-wave interactions in orbitally active quasi-two-dimensional Fermi gases. *Phys. Rev. A* 110, L051302 (2024)
- K. G. Jackson, C. J. Dale, J. Maki, **K. G. S. Xie**, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent s-wave interactions between identical fermions in quasi-one-dimensional geometries. *Phys. Rev. X* 13, 021013 (2023)
- C. R. C. Buhariwalla, Q. Ma, L. DeBeer-Schmitt, **K. G. S. Xie**, D. Pomaranski, J. Gaudet, T. J. Munsie, H. A. Dabkowska, J. B. Kycia, and B. D. Gaulin. Long-wavelength correlations in ferromagnetic titanate pyrochlores as revealed by small-angle neutron scattering. *Phys. Rev. B* 97, 224401 (2018)

IN REVIEW

- K. G. S. Xie***, C. J. Dale*, K. Pond Grehan, M. F. Wang, T. Enss, P. S. Julianne, Z. Yu, and J. H. Thywissen. Dimer-projection contact and the clock shift of a unitary Fermi gas. (under review)

Presentations

CONTRIBUTED TALKS AND POSTERS

- K. G. S. Xie**, C. J. Dale, K. Pond Grehan, M. F. Wang, Z. Yu, P. S. Julianne, T. Enss, and J. H. Thywissen. Clock shift and contact of a unitary Fermi gas inferred by dimer association. Oral presentation: DAMOP 2025, Portland, Oregon, U.S.A.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent s-wave dimers near a p -wave Feshbach resonance in a strongly confined Fermi gas. Oral presentation and poster: ICPEAC 2023, Ottawa, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, S. Zhang, and J. H. Thywissen. Associating emergent s-wave dimers along strongly confined directions of a spin-polarized Fermi gas. Oral presentation: DAMOP 2023, Spokane, Washington, U.S.A.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Feshbach-enhanced odd- and even-wave interactions between identical fermions in quasi-one-dimensional confinement. Poster: ICAP 2022, Toronto, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Kokkelmans, S. Smale, P. S. Julianne, and J. H. Thywissen. P -wave interactions in a spin-polarized Fermi gas with and without confinement. Poster: DAMOP 2022, Orlando, Florida, U.S.A.

Honors & Awards

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| 2025 | SGS Conference Travel Grant , University of Toronto | <i>Toronto, Canada</i> |
| 2023 | Ontario Graduate Scholarship , University of Toronto | <i>Toronto, Canada</i> |
| 2021 | Canada Graduate Research Scholarship – Master’s , NSERC | <i>Toronto, Canada</i> |
| 2020 | Dean’s Honours List , University of Waterloo | <i>Waterloo, Canada</i> |
| 2017 | Undergraduate Student Research Award , NSERC | <i>Waterloo, Canada</i> |
| 2016 | President’s Scholarship of Distinction , University of Waterloo | <i>Waterloo, Canada</i> |

Teaching Experience

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| 2026 | PHY385 , Introductory Optics, Teaching Assistant | <i>Toronto, Canada</i> |
| 2025 | PHY324 , Practical Physics II, Teaching Assistant | <i>Toronto, Canada</i> |
| 2025 | PHY293 , Waves and Modern Physics, Teaching Assistant | <i>Toronto, Canada</i> |
| 2024 | PHY294 , Quantum and Thermal Physics, Teaching Assistant | <i>Toronto, Canada</i> |
| 2024 | PHY405 , Electronics, Teaching Assistant | <i>Toronto, Canada</i> |
| 2023 | PHY424 , Advanced Physics Laboratory, Teaching Assistant | <i>Toronto, Canada</i> |
| 2021, 2022 | PHY224 , Practical Physics I, Teaching Assistant | <i>Toronto, Canada</i> |
| 2020, 2021 | PHY131 , Introduction to Physics I, Teaching Assistant | <i>Toronto, Canada</i> |